

(1)

Shortest Path Problem:

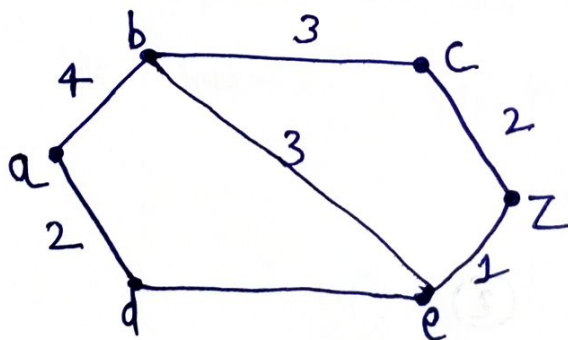
Let $G = (V, E)$ be a weighted graph. To find the shortest path distance between the two specific vertices of G is known as the shortest path problem.

Ans.

Dijkstra's Algorithm:

This algorithm proceeds by finding the length of the shortest path from 'a' to the first vertex, the length of the shortest path from 'a' to the second vertex, and so on, until the length of the shortest path from 'a' to z is found.

Ex: What is the shortest path between a to z in the weighted graph shown in the following figure.



①
Soln: For the shortest path, the only path beginning at 'a' that contain no vertex other than 'a' are (a, b) and (a, d) . As their lengths are 4 and 2 respectively, it follows that 'd' is closest vertex to a.

We can find the next closest vertex by looking at all path that go through only a and d is (a, d, b) & (a, d, e) , and the shortest path to e is (a, d, e) with length 5.

To find the next closest vertex 'a', we want to examine the only path go through only a, d, and e. so consequently z is the next closest vertex to a, and the length of the shortest path to z is 6.



Details of Dijkstra's Algorithm:

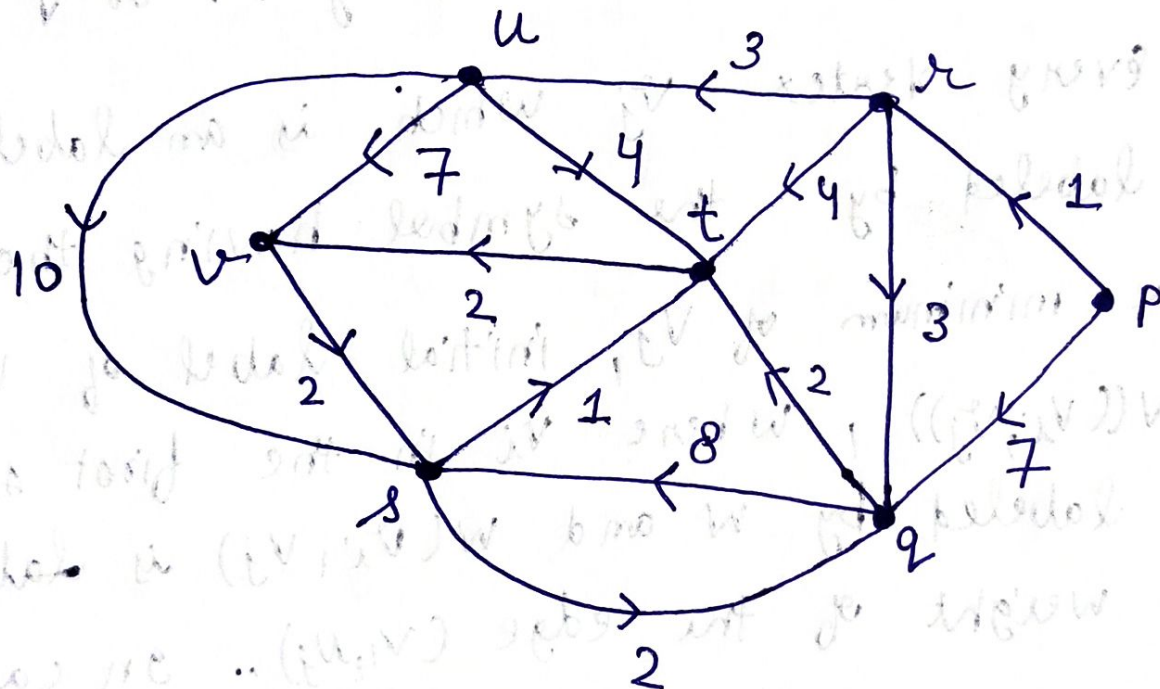
In this algorithm, following a weighted graph the initial vertex is temporarily labeled (marked) with 0 and the other vertices with a permanent label ∞ . In the first step, the other vertex, under the following rules, is labeled by $\checkmark$ or $\vee$

(i) Every vertex v_j which is already labeled, is labeled by the symbol having the value minimum of v_j , initial label of v_j + $W(v_i, v_j)$, where v_i in the first step is labeled by 0 and $W(v_i, v_j)$ is labeled the weight of the edge (v_i, v_j) . In case there is no edge between v_i and v_j then we have $W(v_i, v_j) = \infty$.

(ii) Now we find the minimum value of all the vertex labeled by ∞ , if there is tie between, for the minimum value

of any two vertices, then no vertex can be selected. The iteration takes place until we reach the destination vertex.

Q1) What is the length of the shortest path between the vertices 'p' and 'v' in the following weighted graph.



Soln :

	p	q	r	s	t	u	v
0	✓	∞	∞	∞	∞	∞	∞
0	7	1	∞	∞	∞	∞	
0	7	1	✓	∞	∞	∞	
0	4	1	∞	5	4	∞	
0	4	1	14	5	4	11	
0	4	1	14	5	4	11	
0	4	1	12	5	4	11	
0	4	1	12	5	✓ 4	11	
0	4	1	12	5	4	7	
0	4	1	12	5	4	7	✓

The shortest path is

$p \rightarrow r \rightarrow t \rightarrow v$.

the length of shortest path between the vertices p and v is —

5

(5)